

Monolithic Spring Boot Application

GitHub README / Project Documentation

1. Project Overview

This project is a monolithic Spring Boot application developed by combining the main business functionalities of a microservices-based application into a single application. The application contains User, Product, and Order modules that run together as one Spring Boot application.

2. Architecture

The application follows a layered monolithic architecture. All modules are maintained inside one Spring Boot project and share the same application runtime and database.

Layer / Module	Purpose
Controller	Handles REST API requests and responses.
Service	Contains business logic.
Repository	Handles database operations using Spring Data JPA.
Entity / Model	Represents application data such as User, Product, and Order.
Database	Stores users, products, and orders.

3. Main Modules

- **User Module:** Create, read, update, and delete user information.
- **Product Module:** Manage product information including product name, quantity, and price.
- **Order Module:** Manage order-related operations using the user and product information.

4. Technologies Used

- Java
- Spring Boot
- Spring Web / REST APIs
- Spring Data JPA
- Maven
- MySQL
- Postman for API testing
- Git and GitHub for version control

5. Project Structure

```
demo/
■■■■ src/main/java/com/example/demo/
■ ■■■■ controller/
■ ■■■■ service/
■ ■■■■ repository/
■ ■■■■ entity/ or model/
■ ■■■■ DemoApplication.java
■■■■ src/main/resources/
■ ■■■■ application.properties
■■■■ pom.xml
■■■■ README.md
```

6. REST API Testing

Module	Typical Operations	Testing Tool
User	POST, GET, PUT, DELETE	Postman
Product	POST, GET, PUT, DELETE	Postman
Order	POST, GET, PUT, DELETE	Postman

7. How to Run the Project

1. Clone or download the project from GitHub.
2. Open the project in VS Code or another Java IDE.
3. Make sure Java and Maven are installed.
4. Configure the database connection in application.properties.
5. Open the terminal in the project root folder.
6. Run: `mvn spring-boot:run`
7. Wait until Spring Boot starts successfully.
8. Use Postman to test the User, Product, and Order REST APIs.

8. Example Maven Command

```
mvn spring-boot:run
```

9. Advantages of the Monolithic Architecture

- Simple project structure and deployment.
- All business modules are available in one application.
- Easy to develop and test for small or medium-sized applications.
- Straightforward communication between modules.
- Single application deployment.

10. Microservices to Monolith Conversion

The original application was designed around separate business services. In this version, the User, Product, and Order functionalities have been organized into one Spring Boot application. This

demonstrates how the same business functionality can be implemented using a monolithic architecture.

11. GitHub

The project source code and related project documentation can be maintained in the GitHub repository. The repository should contain the application source code, Maven configuration, documentation, and other necessary project files. Generated build files such as the target directory should normally be excluded using `.gitignore`.

12. Conclusion

This project demonstrates a complete monolithic Spring Boot application with User, Product, and Order modules. REST APIs can be tested using Postman, while Git and GitHub are used for source-code management and project sharing.